

Worksheet 3

Problem 1. Polynomial Spaces & Linear Independence

Consider the vector space $V = \mathcal{P}_3(\mathbb{R})$ of polynomials with real coefficients of degree at most 3. Let $S = \{p_1(x), p_2(x), p_3(x), p_4(x)\}$ be a subset of V , where:

$$\begin{aligned}p_1(x) &= 1 + 2x - x^2 \\p_2(x) &= 3 + x^2 + x^3 \\p_3(x) &= 2 - 2x + 2x^2 + x^3 \\p_4(x) &= 4 + 4x - x^2 + x^3\end{aligned}$$

- (a) Determine whether S is linearly independent or linearly dependent.
- (b) Find a basis for $\text{Span}(S)$ and state its dimension.
- (c) Extend the basis found in part (b) to a basis for all of $\mathcal{P}_3(\mathbb{R})$.

Problem 2. Subspaces & Dimension

Let W be the subset of $\mathcal{P}_2(\mathbb{R})$ defined by:

$$W = \{p(x) \in \mathcal{P}_2(\mathbb{R}) \mid p(1) = 0 \text{ and } p'(0) = 0\}$$

- (a) Prove that W is a subspace of $\mathcal{P}_2(\mathbb{R})$.
- (b) Find a basis for W and determine $\dim(W)$.
- (c) Construct a subspace $U \subset \mathcal{P}_2(\mathbb{R})$ such that $\mathcal{P}_2(\mathbb{R}) = W \oplus U$ (a direct sum).

Problem 3. Linear Maps on Polynomial Spaces

Define a transformation $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ by:

$$T(p(x)) = x \cdot p'(x) + p''(x)$$

- (a) Prove that T is a linear map.
- (b) Find the matrix representation $[T]_B$ of T relative to the standard basis $B = \{1, x, x^2\}$.
- (c) Determine the nullity ($\dim(\ker(T))$) and rank ($\dim(\text{im}(T))$), and state whether T is invertible.